

# MC103 - Cryptography and Blockchain

## Exam

Kristiania University  
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### Notes

It is suggested to pay close attention to the instructions and to use software provided below to ease solving some of the tasks given in the exam.

- <https://www.openssl.org/>
- <https://gnupg.org/>
- <https://www.cryptool.org/en/cto/>

It is suggested to use scripting and python to crack some of the “puzzles” below, that require brute-force search. When using code, provide the Git repository link to your solutions.

1. (6pts) You have been given the following cypher texts. You have a strong indication that they have been encoded by a CAESAR cypher.  
Decipher the following texts to retrieve the original messages. Provide the plain text and explain how you cracked it, and what assumptions have you made.
  - IJKLE LWPZQ LRTCW QFWWZ QDZFY OLYOQ FCJDT RYTQJ TYRYZ ESTYR
  - QBBJXUMEHBTYIQIJWUQDTQBBJXUCUDQDTMECUDCUHUBOFBQOUHI
  - ZA z9 TcVmVi A5 3zE kyz4x9 r SzA sF vE6r4uz4x R26yrsvA r4u ruuz4x 4B3sv89

2. (10pts) You have been given the following cypher text. Hint: the cipher used to encrypt the plaintext is one of the tools available on the webpage:  
<https://www.cryptool.org/en/cto/>

Wkf#rvj`h#aqltm#el{#ivnsp#lufq#wkf#obyz#gld#bmg#ab`h-

- (1pt) Does it look as if a substitution or transposition cipher has been used? Why?
- (3pts) What are the possible cyphers you should consider to decipher it?
- (6pts) Try to decipher it, assuming that the key is within the first FOUR most obvious keys (or most systematic ways of searching for the key). If you have tried FOUR or more keys for a given cipher it is likely NOT the cipher you need. Explain your process.

3. (10pts) Consider the text below. Same hint as above.

Teu cgeno jmsvr hlz dghqi kref xup oete ayo.

- (1pt) Does it look as a substitution or transposition cipher has been used? Why?
- (3pts) What are the possible ciphers you should consider to decipher it?
- (6pts) Try to decipher it, assuming that the key is within the first FOUR most obvious keys (or most systematic ways of searching for the key). If you have tried FOUR or more keys for a given cipher it is likely NOT the cipher you need. Explain your search process.

4. (10pts) Given those four names: Mariusz Sarah Fabian and Daniel

Find and answer the following:

a) Which name hashes to

3b472fb496e7185001072f414b34f91b252a982e

with SHA-1 algorithm?

b) Which name hashes to

262031397020fd8df478ec13b4b096c5

with MD5 algorithm?

c) What is the hash SHA256 of the name: Mariusz

d) What is the hash RIPEMD-160 of the name: Fabian

e) What is the length in BITS of the following hash? Why? Explain how you can calculate it. What digest (hashing) algorithms do you know with that length?

408d890298c4963dfaacc2e8b508552b9e1ee8048

**NOTE:** pay attention to the “end-of-line” character. Hash of “Mariusz” is different to “Mariusz\n”

5. (10pts) PGP and SSH keys

a) What is the algorithm used in the public PGP key, and how did you checked the algorithm?

Note: the pgp keys are in External Resources on Canvas.

- pgp\_fabian\_pub.asc
- pgp\_mariusz\_gmail.asc

b) Demonstrate that you can use SSH keys to login to SSH server, and to commit to Git repository without the need to provide password (proper setup of SSH keys with ssh-agent working)

c) Demonstrate that you can send encrypted messages via email.

d) Demonstrate that you can send signed messages via email.

e) Demonstrate that you can verify signature of a software package downloaded from Internet, eg. for Bitcoin Core client - <https://bitcoin.org/en/download>

6. (4pts) Briefly compare PGP keys generated with RSA3072 and with ED25519. Explain their characteristics such as length, and pros and cons as a short bullet-list comparison. Can you quickly visually distinguish one from the other?

7. (10pts) In the course of your investigations, you have managed to obtain the private PGP key of a suspect. The private key file is called: **keyfile-secret** and shared on the course shared drive. Here is a copy:

```
-----BEGIN PGP PRIVATE KEY BLOCK-----

lIYEZ8CSRYJKwYBBAHARw8BAQdAI0dxd34CXMubQwXqQLBQb0cz7ZzMfNjK7iG+
5mMLbO3+BwMC2KAE1gQrm4f2Kbec3mtU3qk0RpfSOXfMXpFOLz79V6/9JO1NXeO7
KAfW9S8qs/j/KKWfM5gKvK4F7P3SpL+sdhXARwPAgg2Upa8M3qu0e7QnU2FudGEg
Q2xhdXMgPHNhbnRhLmNsYXVzQG1pY3Jvc29mdC5jb20+iJkEEYKAEWIIQQLnCVw
unbeY+DAQs+wIlAKYaQcBwUCZ8CSQIbAwUJAHanAAULCQgHAgIiAgYVCgkICwIE
FgIDAQIEBwIXgAAKCRCwIlAKYaQcB38WAP9oDiOJqjjWiHTGcbKXcbnj7HCUqHdM
i6kR7cc4a5KLFWEA0EijV2DEZDuRCy9gdEiQoZ7zP/wOH+mlZfU2JDc8gAeciwRl
nwJJEgorBgEEAZdVAQUBAQdAlRAujLPSPcOGKtL8nuzkF6bSBBegevMEHZ4+VtEZ6
mGIDAQgH/gcDAs04JlKMOLj/9mGSPYbMioInenKSHlQQHy4P2fBloDNGEY5M90n8
XIhBO5q5uYpaoCKrZ9lqNaYvnY1i56QM5jFLHPR9G+xO4OecH7jtAAOIgfQYFgoA
JhYhBAucJXC6dt5j4MBCz7AiUAphpBwHBQJlnwJJAhsMBQkAdqcAAAoJELAiUAph
pBwHo8IBANrFeVFMCC5/ixWo9klhgRxKJQr7U4rr0T4BkvU0VK+8APsEKsHmadO4
7WS0awZDL9iQCillBrDHLtgSCrN6J+1KCw==
=Jx+r
-----END PGP PRIVATE KEY BLOCK-----
```

You have also successfully guessed the password for the file, and it is “santa”. Can you explain how you have possibly guessed that?

You have also obtained a secret message encrypted with this key. The message is called **msg.txt.asc** and it is also shared in the shared course drive. Decrypt the message using the provided secret key. Explain how you have achieved it, step by step.

```
-----BEGIN PGP MESSAGE-----

hF4D9TO5rerfDDcSAQdAZSBCwPrcVaqhccxGA8cFo9K1lxjICpJaIYapLYM153Ew
+dBjTBLS3VmH/cckUTsnLBb9IRYbrJwooQDyM6GNF1YFjGSfuCgBAm9HsgljMtEC
1IUBCQIQDIJcDhOf9mJzpsltoWtajJpy+88DNLws98e5nPYrEfD+kfo4UcNIL6yP
p3jw+P0UuURPEG2tiXd73bU141tDeYoyKwt01iaWGxTnFjts83vD2S2zNXteQiQ5
txQpnFpF6imDjQqAtHFI8X1LTpxLHvNRv2D2n+R97B9sNmkliQ9B
=XbrY
-----END PGP MESSAGE-----
```

8. (6pts) Bitcoin - answer questions and document how you obtained the answers:

- a) How many bitcoins have been seized in the Silk Road case, and what has been the official FBI address used for the seizer?
- b) If by a programmer error, a public key 0 is picked to generate a public-private key pair, what would be the Bitcoin address for that public key? When was the last transaction and what is the status of UTXOs with this address? Explain your sources.
- c) How many transactions block 157,000 in Bitcoin has? How many bitcoins have been spent?
- d) What this represents? Who was involved?

f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16

- e) Explain the transaction, what has been transferred to whom:

cc455ae816e6cdafdb58d54e35d4f46d860047458eacf1c7405dc634631c570d

- f) Explain the transaction. How much money has changed hands? What this transaction represents?

74b164edf6f40a00c61304a234d01e8263c04bcc57811aab595ffaeb2468f81a

9. (10pts) Ethereum

- a) What is the address `0x7da82C7AB4771ff031b66538D2fB9b0B047f6CF9` for? Is it an Externally Owned Account or Contract address? People/project from which country ultimately own the funds?
- b) When and by whom was the contract created, and what does it do?

0xCA9Df33e48f88259ef2d06a790b13Db2ABBDf0C1

- c) What is this contract for?

0xb541fc07bc7619fD4062A54d96268525cBC6FfEF

10. (10pts) Using Metamask wallet together with `hardhat` development network, deploy two standard ERC-20 tokens on Sepolia network. This is exactly the same task as the homework about ERC-20 tokens. Follow the same rules as the homework from week 1. You can re-use the token created for the homework.

11. (15pts) Exchange contract: implement a Token Swap Mechanism

### Concept: Fixed-Precision Arithmetic in Solidity

Solidity does not support floating-point numbers, so you must store the exchange rate as a ratio between two assets. Be mindful of **precision loss** and **integer overflows**.

## Tasks

- Create a smart contract that stores an exchange rate between two ERC-20 tokens as a ratio.
- Implement the exchange function:

```
function exchange(uint256 asset1) external returns (uint256);
```

- Converts `asset1` into `asset2` based on the stored exchange rate
  - Emits an event when a swap occurs.
- Precision & Security Considerations
  - Implement fixed-precision arithmetic (e.g., multiplying first, then dividing)
  - Prevent **overflows** by using **SafeMath** or Solidity's built-in checks - use `unchecked` blocks where appropriate
- Testing & Deployment
  - Ensure Solidity compiler does not produce any errors or warnings.
  - Deploy and test the contract in Remix or using the command line with hardhat.
  - Demonstrate a **successful token swap** between your ERC-20 token and the other token you deployed earlier using your Metamask wallet and etherscan explorer
- Optional (7pts) Web interface
  - Implement the Exchange interface as a simple front-end webapp. The webapp should connect to the user's Metamask wallet, and allow the user to exchange Coin1 to Coin2, or Coin2 to Coin1 based on the exchange rate.
  - It should allow the owner to update the exchange rate.

## Deliverables

1. Your wallet address(es), and addresses of your contracts for ERC-20 tokens. Deployed ERC-20 tokens must be visible on Sepolia Etherscan and in your Metamask wallet.
2. Address of your Exchange contract
3. TXs hashes demonstrating a functional behaviour of your Exchange contract with event emissions.
4. Explanation of how precision loss and overflow were handled in the exchange function.
5. Git repository with your project files.

Try to optimize the exchange function to minimize gas costs while maintaining accuracy and security.

This assignment will help you build real-world smart contracts, preparing you for decentralized finance (DeFi) applications. Happy coding!

## 12. (7pts) IPFS

- Demonstrate that you can add and publish globally a file into the IPFS network. The file hash should be accessible via a public gateway.
- Demonstrate that you can run an IPFS node on your computer behind a firewall and tunnel the node traffic through a public IP address of a remote server with public IP address. That

means, demonstrate that a public IPFS gateway can read a locally pinned file from your IPFS node on your laptop from behind a firewall.